

Efstathia Soufleri

Machine Learning Researcher · Trustworthy AI · LLM Evaluation · Efficient Deep Learning

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Professional Summary

Machine learning researcher with a Ph.D. from Purdue University and two years of postdoctoral experience. Develops privacy auditing methods, differentially private generative models, LLM evaluation pipelines, and efficient vision systems in Python and PyTorch. Research published at NeurIPS, ICML, ICLR, CVPR Workshops, ACL, EMNLP, and TMLR, including a NeurIPS 2024 Spotlight. Experienced in research prototyping, benchmark development, and mentoring researchers.

Technical Skills

Programming and tools: Python, PyTorch, C, C++, Bash, Linux, Docker, MATLAB.

Trustworthy AI and generative ML: Differential privacy, membership inference, memorization estimation, out-of-distribution detection, generative modeling.

LLMs and efficient ML: Large language models (LLMs), natural language processing (NLP), multimodal learning, quantization, model compression, knowledge distillation, early exits.

Frameworks and evaluation: TensorFlow, Keras, MXNet; benchmark development, training and evaluation pipelines, hardware-software co-design.

Professional Experience

Postdoctoral Researcher | Sep 2024 - Aug 2026 Archimedes AI Research Unit, Athena Research and Innovation Center, Greece

- **Privacy auditing:** Developed fast, formal memorization estimators (ICML 2025) and sample-gradient-based memorization proxies (ICLR 2026).
- **LLM evaluation:** Built evaluation pipelines for Greek finance and multilingual, multimodal financial benchmarks (EMNLP 2025; ACL 2026).
- **Robustness:** Co-developed out-of-distribution detection using additive angular margin loss (CVPR Workshops 2025).
- **Research mentorship:** Co-supervised two projects resulting in publications on differentially private mental-health classification (Frontiers in Digital Health, 2025) and cross-modal generative augmentation for biological classification (TMLR, 2026).

Graduate Research Assistant | Aug 2017 - Aug 2024 Purdue University, USA | C-BRIC and COCOSYS | Advisor: Prof. Kaushik Roy

- **Private image synthesis:** Designed DP-ImgSyn, a discriminative framework with formal differential privacy guarantees; evaluated synthetic data using downstream classification (TMLR 2024).
- **Privacy attacks:** Co-developed curvature-based methods linking loss geometry to membership inference and memorization (NeurIPS 2024 Spotlight; ICML 2024).
- **Efficient video recognition:** Developed compressed-video action recognition pipelines with progressive knowledge distillation and early exits.
- **Model compression:** Built a PyTorch framework for automatic layer-wise mixed-precision quantization, reducing network size by up to 6 times.
- **Hardware efficiency:** Developed a hybrid RRAM-SRAM system, reducing energy requirements by up to 2.88 times while maintaining vision-task accuracy.

Professional Experience Continued

Graduate Research Assistant | Mar 2016 - Jul 2017 Department of Computer Science, University of Thessaly, Greece

- Developed video compression heuristics and matrix partitioning algorithms for real-time multimedia processing.

Education

Ph.D., Electrical and Computer Engineering | Aug 2024 Purdue University, USA | GPA: 3.83/4.00 Dissertation: Privacy Preserving and Efficient Machine Learning Algorithms

M.Sc., Computer Science | Jul 2017 University of Thessaly, Greece | GPA: 9.72/10.00 | Valedictorian

B.Sc., Mathematics | Feb 2016 National and Kapodistrian University of Athens, Greece | Applied Mathematics

Selected Publications

Full publication list: efstathia-soufleri.github.io

Memorization Through the Lens of Sample Gradients - ICLR 2026

Ravikumar, D., Soufleri, E., Hashemi, A., and Roy, K.

Towards Memorization Estimation: Fast, Formal and Free - ICML 2025

Ravikumar, D., Soufleri, E., Hashemi, A., and Roy, K.

Curvature Clues: Decoding Deep Learning Privacy with Input Loss Curvature - NeurIPS 2024, Spotlight

Ravikumar, D., Soufleri, E., and Roy, K.

DP-ImgSyn: Dataset Alignment for Obfuscated, Differentially Private Image Synthesis - TMLR 2024

Soufleri, E., Ravikumar, D., and Roy, K.

Plutus: Benchmarking Large Language Models in Low-Resource Greek Finance - EMNLP 2025

Peng, X., Papadopoulos, T., Soufleri, E., et al.

MultiFinBen: Benchmarking Large Language Models for Multilingual and Multimodal Financial Application - ACL 2026

Peng, X., Qian, L., Wang, Y., et al. (including Soufleri, E.).

Learning Unified Spatio-temporal Representations for Efficient Compressed Video Understanding - WACV 2026

Das Biswas, S., Soufleri, E., Roy, A., and Roy, K.

Leadership and Recognition

- **Research service:** TMLR Action Editor; reviewer for NeurIPS, ICML, ICLR, CVPR, TMLR, and other ML venues. Top Reviewer, NeurIPS 2025; Gold Reviewer, ICML 2026.

- **Teaching and outreach:** Delivered preparatory lectures for the Greek IOAI 2026 team and accompanied the team as official Observer in Astana.

- **Invited talks:** University of Manchester (2025); Athena Research and Innovation Center (2025).

- **Award:** Gerondelis Foundation Award for Academic Excellence (2021).

Languages

Greek: Native | English: Full professional proficiency | French: Professional working proficiency